

D EESHAN PAVAN KARTEEK

Ph +91-81066 17544

Mail : eeshankarteek26@gmail.com

<https://github.com/lameeshan26>

<https://www.linkedin.com/in/dontarajueeshan/>

Summary

ML-focused Computer Science student with experience building deep learning models and real-time optimization systems. Demonstrated ability to improve model accuracy and system efficiency through data-driven approaches. Skilled in Python, TensorFlow, and scalable ML pipelines, with working knowledge of Java for backend development.

SKILLS

Languages: Python, Java

Machine Learning: TensorFlow, Scikit-learn, XGBoost, CNN Architectures

Backend Development: Flask, REST APIs, Node.js

Data & Databases: MySQL, Firebase, Pandas

Systems & Tools: Kafka (event-driven systems), Git, GitHub

Development Tools: Jupyter Notebook, Google Colab

Internship

Gen AI Intern – Smart Bridge

Dec 2025 – Mar 2026

- Developed a Generative AI application using Gemini API supporting text and image-based inputs.
- Engineered an end-to-end pipeline for input processing, prompt generation, and response handling.
- Automated recipe generation and nutritional analysis, reducing manual search time by ~70%.
- Integrated image-based input processing to extract food information and generate contextual outputs.
- Processed 20+ user inputs / test cases for prompt validation.

PROJECTS

Bone Fracture Classification Model: Python, TensorFlow, Flask, Scikit-Learn, MySQL, VGG19, EfficientNet, PIL

- Engineered and benchmarked multiple CNN architectures (MobileNet, VGG19, EfficientNet, DenseNet) for fracture classification.
- Improved model accuracy from 84.2% → 92.4% (+8.2%) by designing a hybrid architecture (DenseNet121 + XGBoost).
- Benchmarked deep learning + classical ML combinations (SVM, Random Forest, XGBoost) to optimize performance.
- Selected final model based on highest validation accuracy and generalization performance.
- Deployed model via Flask API for real-time inference.
- Trained on 10,000+ labeled X-ray images across 10 classes.
- Evaluated using precision, recall, and F1-score for robust performance validation.

Last-Mile Logistics Optimizer: Python, OR-Tools, XGBoost, Kafka, TomTom API, PostGIS, Pandas

- Built a real-time, event-driven route optimization system integrating live traffic and weather data.
- Reduced fuel costs by 20% and delivery time by 15% through traffic-aware routing and speed modeling.
- Improved ETA prediction accuracy by 25% using ML-based dwell time estimation.
- Enabled real-time route re-optimization with sub-second response using Kafka-based event triggers.
- Increased driver capacity by 10% through dynamic re-routing.
- Architected a hybrid ML + optimization pipeline to predict hidden delays (parking, congestion patterns).
- Evaluated across 300+ simulated delivery routes using real-world traffic data.

EDUCATION

B-Tech,CSE(AI&ML), Sri Venkateshwara College of Engineering, CGPA : 8.74

2022 - 2026